

WASP-36b

R-1135

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_220227 · created 2026-07-30T23:31:11+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459637.715 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.166 +/- 0.049
Transit depth (Rp/Rs)^2	2.75 +/- 1.61 [%]
Orbital Inclination (inc)	87.56 +/- 2.87
Airmass coefficient 1 (a1)	0.6 +/- 0.4
Airmass coefficient 2 (a2)	0.43 +/- 0.72
Scatter in the residuals of the lightcurve fit is	73.74 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	6.75
Transit Duration (day)	0.0924 +/- 0.0093

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.166 ± 0.049 vs 0.1368 (deviation 0.6σ)
Duration fit vs published	0.0924 d vs 0.0758 d (deviation 1.78σ)
Mid-time O–C	11.1 min
Depth SNR	3.4
Residual scatter	73.74 %

Light curve

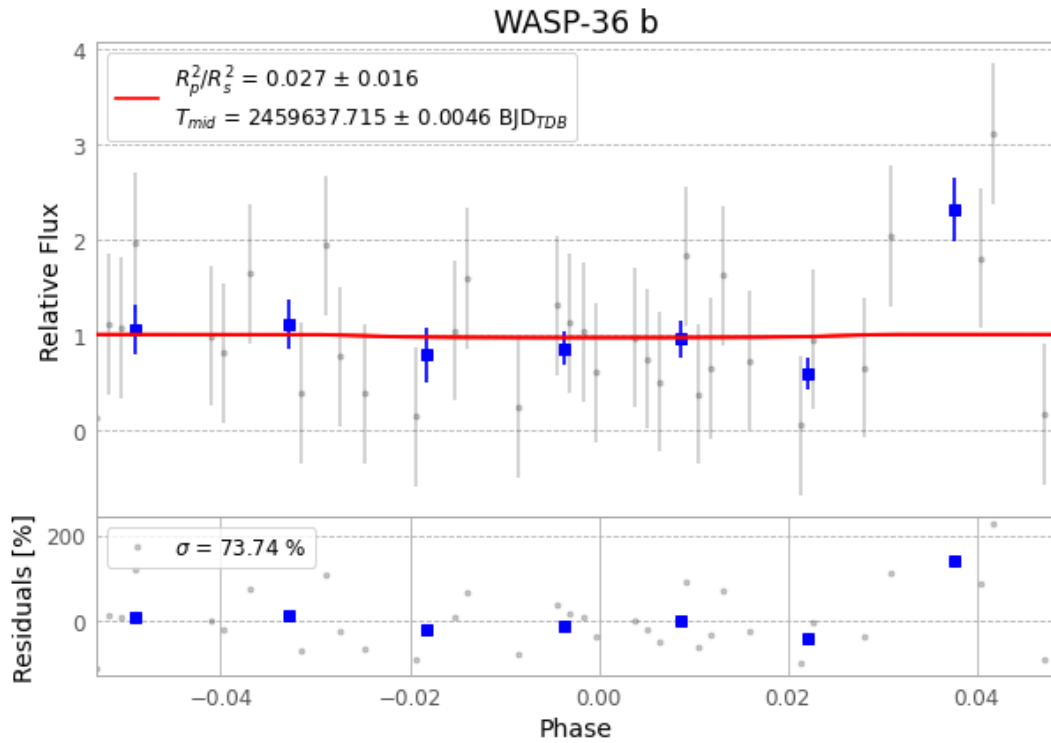


Figure 1 — FinalLightCurve_WASP-36 b_2022-02-26.png (SHA-256 1839b694302f1bd4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [410, 450], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c37c3651009189b1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36220227030012.FITS → WASP-36220227064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-220227.log (44d2678b6b5e...), FinalLightCurve_WASP-36 b_2022-02-26.png (1839b694302f...), FinalLightCurve_WASP-36 b_2022-02-26.csv (adbdb2d376be...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:31Z.